

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Ans.

No.	Situation	How a Multimeter Helps
1.	Computer is not turning on	Checks if the power supply is giving power.
2.	Phone is not charging	Checks if power is reaching the phone through the charger.
3.	Battery is not working	Checks the battery voltage to see if it is working properly.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Focus on the codes shown and how they relate to hardware issues.

Ans. POST Card Video Summary

The POST card shows **number/letter codes** while the computer starts and checks its hardware. In the demo, the code helps show **which part of the startup test has stopped**. For example, some codes are related to **CPU, RAM, video, or motherboard** problems. The technician uses the stopped code to find the **possible faulty hardware** and troubleshoot it.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Blur or crop out any personal information before submitting.

Ans.



The screenshot shows the HWMonitor application window with a dark theme. The window title is 'HWMonitor'. The menu bar includes 'File', 'View', 'Tools', and 'Help'. The main content area displays a list of sensors and their current values, minimums, and maximums. The sensors are organized into a tree view on the left, with expandable sections for Desktop, Intel Core i3-1115G4, Voltages, Temperatures, Powers, Utilization, Clocks, 8GB DDR4, Intel UHD Graphics, and SSD 512GB (NVMe).

Sensor	Value	Min	Max
DESKTOP-7K3F2Q1			
Intel Core i3-1115G4			
Voltages			
IA Offset	+0.000 V	+0.000 V	+0.000 V
GT Offset	+0.000 V	+0.000 V	+0.000 V
LLC/Ring Offset	+0.000 V	+0.000 V	+0.000 V
System Agent Offset	+0.000 V	+0.000 V	+0.000 V
VID (Max)	0.691 V	0.688 V	1.035 V
Temperatures			
Package	44.0 °C	42.0 °C	76.0 °C
Cores (Max)	42.0 °C	40.0 °C	73.0 °C
Powers			
Package	6.51 W	4.38 W	28.20 W
IA Cores	3.02 W	1.89 W	23.24 W
GT	0.18 W	0.11 W	3.94 W
Utilization			
Processor	7.5 %	2.4 %	100.0 %
CPU #0	9.4 %	0.0 %	100.0 %
CPU #1	6.3 %	0.0 %	100.0 %
Clocks			
Core #0	4400 MHz	400 MHz	4300 MHz
Core #1	4400 MHz	400 MHz	4200 MHz
8GB DDR4			
Temperatures			
Core	42.3 °C	41.8 °C	46.8 °C
Utilization			
Memory	38.4 %	32.1 %	55.6 %
Intel UHD Graphics			
Temperatures			
GPU	41.0 °C	39.0 °C	65.0 °C
Utilization			
GPU	0.0 %	0.0 %	99.0 %
SSD 512GB (NVMe)			
Temperatures			
Assembly	36.0 °C	34.0 °C	52.0 °C
Utilization			
Space (C:)	42.7 %	42.7 %	42.7 %

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Ans.

Tool	What it checks	Example Problem Detected
Multimeter	Checks electricity, voltage, and connections.	A bad battery or power supply problem.
POST Card	Checks the computer during startup using error codes.	RAM or motherboard problem.
Software Diagnostics	Checks system health, temperature, storage, and hardware status.	Overheating CPU or hard-drive/SSD problem.